

Adarsh Raj

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Summary

Data and AI Engineer with enterprise-scale experience designing and delivering intelligent data platforms, semantic search systems, FastAPI microservices, and Spark-based pipelines. Proven track record of eliminating manual effort through automation, improving data retrieval performance, and owning end-to-end DevOps. Skilled across the full stack: AI/ML integrations, backend engineering, cloud infrastructure, and enterprise system integrations (ServiceNow, Collibra, SailPoint).

Work Experience

Technical Associate — Data & AI Engineering

Sep 2024 – Present

GSK (GlaxoSmithKline), Bengaluru, India

- Semantic Search & AI:** Designed and implemented semantic search and conversational query capabilities for the Rover metadata platform using **OpenAI embeddings** and **Azure AI Search**, enhancing metadata discoverability and user engagement at enterprise scale.
- Spark / Databricks Pipelines:** Owned development and optimisation of **Apache Spark** ingestion pipelines on **Databricks**, achieving significant runtime reductions and enabling automated metadata cataloguing across enterprise systems.
- FastAPI Microservices:** Built and maintained production-grade **FastAPI** microservices for metadata retrieval, data lineage views, and enterprise-wide search with high test coverage and maintainability.
- ServiceNow–Collibra Integration:** Delivered end-to-end governance workflow automation between ServiceNow and Collibra; eliminated **90% of manual effort (~200 hours/year saved)** and achieved a **99.6% reduction in data sync time** (30 min → 7 sec).
- GraphQL Optimisation:** Re-engineered Collibra GraphQL queries, reducing metadata retrieval time by **87.5%** (8 hours → 1 hour).
- SailPoint Integration:** Built SailPoint–Rover integration to streamline access governance, surface entitlement visibility, and simplify permission validation workflows.
- Data Quality Integration:** Led integration of DQ outputs into Collibra, automating registration of DQ metrics and enabling transparent governance reporting across metadata assets.
- DevOps & CI/CD:** Owned end-to-end CI/CD using **GitHub Actions** and **Azure DevOps** – release automation, environment management, and deployment reliability for Metadata Management and GSC DQM products.
- AI / NLP POCs:** Explored NLP-driven analytics in Power BI; evaluated **Databricks Genie** for Natural Language Query (NLQ) feasibility for enterprise adoption.
- Azure Infrastructure:** Designed scalable solutions using **Azure Function Apps**, **Azure Data Factory**, and **Azure SQL** to orchestrate automated data workflows across platforms.

Graduate Intern — Data & DevOps

Jan 2024 – Aug 2024

GSK (GlaxoSmithKline), Bengaluru, India

- Automated reporting workflows using **Azure DevOps**, **Power BI**, and **JIRA**, reducing manual reporting overhead and gaining hands-on project experience across DevOps pipelines and data analytics.

Technical Skills

AI & Search: OpenAI (Embeddings, Chat), Azure AI Search, Semantic Search, NLP, Databricks Genie (NLQ), Power BI NLP

Data Engineering: Apache Spark, Databricks, Azure Data Factory, Azure SQL, Delta Lake, ETL/ELT Pipelines

Backend & APIs: Python, FastAPI, REST APIs, GraphQL, Java

DevOps & Cloud: Azure DevOps, GitHub Actions, CI/CD, Docker, Azure Function Apps, Azure Machine Learning

Enterprise Integrations: Collibra, ServiceNow, SailPoint, Collibra DQ

Other Tools: Power BI, JIRA, SharePoint, Databricks, MySQL

Education

Bachelor of Technology, Computer Science & Information Science

Aug 2020 – May 2024

M.S. Ramaiah University of Applied Sciences, Karnataka, India | CGPA: 8.0/10.0

Relevant Coursework: Database Systems, Artificial Intelligence, Computer Networks, Design Patterns, Operating Systems.

Achievements & Certifications

- Rank **5** on GeeksforGeeks at university level; Rank **281** in GFG Coding Contest 119.
- Best Project Award** out of 250 students at M.S. Ramaiah University.
- Certifications: **SQL** (HackerRank), **Azure DevOps** (Udemy), **Generative AI** (Microsoft), Introduction to Cybersecurity (Cisco), Python Masterclass (Udemy).